

Exercise Sheet

Using LLMs to Programmatically Extract and Curate Research Data

ResBaz NZ | Participant copy

Setup reminder

1. Create workshop folder: Name: Data curation workshop 30 June 2026
 2. Download: FOUR items from Zoom chat `plants_site_A.csv`, `plants_site_B.csv`, `plants_sites_AB_large.csv` and Exercise Sheet
 3. Open Copilot at copilot.microsoft.com
 4. Open Word — title: Data curation workshop prompt log – 30 June 2026
 5. Open Exercise Sheet (tick)
 6. Open `plants_site_A.csv`: Have a look at the data, get your head around it, ask any questions in chat – what might be problematic?
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Segment 1 — Intro + framing

Segment 2 — Block 1: Data cleaning

Research task completion framework

Use these steps for every exercise.

Step		What you do
1	Identify task	Understand what is required
2	Identify potential issues	Look at data, what's between it & completing task?
3	Copilot assist	Run a prompt to fix the issue; re-run until outputs mean task completion
4	Validate outputs	Check the issue is fixed and the task is complete
5	Log it in Word	Note: (a) data affected, (b) prompt, (c) what changed
6 (Optional)	Apply elsewhere	Use the same prompt on other columns, repeat steps 3-5

2.0 Stage 1 — Data exploration

Drag and drop `plants_site_A.csv` into Copilot.

Prompt →

I am a researcher studying the effect of weed management on native plant restoration across two sites in West Auckland. I want to understand whether weeding alongside native planting (Site A) produces better plant outcomes than native planting alone (Site B).

Attached is my data set (`plants_site_A`)

Tell me what you understand about this data – the columns, and what they seem to measure.

(Add this prompt to Word prompt log.)

Stage 2 — Numeric: `Height_cm`

Steps 1-2: Identify

Prompt →

Look at ``Height_cm`` only. Return a table with the mean, min, max, range.

Prompt →

Look at ``Height_cm`` only. Return a two-column table with the original column (``Hieght_cm``), and a new column (``Height_cm_cleaned``) with potential outlier values marked with "check".

Step 3: Fix

Prompt →

Modify value 340 to 34 and return the two-column table, do not include any notes in the new cell, jsut "34".

Step 4: Validate — paste into `plants_site_A`; check first, last, changed rows

Step 5: Log — copy prompt to Word table | column | prompt | summary of changes |

Step 6: Apply

Prompt →

Provide me the same summary stats and outlier checks for ``Num_Leaves`` and ``Num_Flowers``.

Prompt →

```
Modify values and return four column table (original column + _cleaned):  
`Num_Leaves`: 847 -> 87.  
`Num_Flowers`: -3 -> 3. Keep 22 as 22.
```

2.3 Categorical: `Site`, `Location`, `Species`

Steps 1-2: Identify — Excel: Data → Filter, check `Species` dropdown

Step 3: Fix

Prompt →

```
Clean categorical columns `Site`, `Location`, `Quadrat`, `Species` and  
`Flowering`.  
I want them to be consistently spelt, all values lower-case, and values  
separated by underscores "_".  
  
Return pairs of columns for each variable: the original and a new column  
`[column_name]_cleaned` for  
each. In the cleaned column have the new, corrected value and use "check" if you  
are unsure.
```

Step 3b (optional): Condition

Prompt →

```
For `Condition`, return the original column and a new column `Condition_cleaned`  
with suggested categories representing a consistent five-point scale from poor  
to excellent.
```

Step 5: Log — copy prompts to Word

2.4 Block 1 close

Segment 3 — Block 2: Feature extraction

3.1 Plant_Age_Months — guided

Prompt →

```
Look at `Date_Planted` and `Date_Measured`. These columns contain dates  
in inconsistent formats – some DD/MM/YYYY, some D/M/YY, some DD-Mon  
with no year, and possibly some US-format (M/D/YY).  
  
Without standardising the formats first, calculate the age of each plant  
in whole months at the time of measurement. Add this as a new column  
called `Plant_Age_Months`. Return these three columns as a table.  
Flag any row where you cannot confidently interpret the dates with "check".
```

Step 4: Validate — paste into spreadsheet; check row count, spot-check 3 rows, flag any negatives

Step 5: Log

3.2 Plant_ID — your turn

Task: Create a `Plant_ID` column — a unique identifier built from Site, Location, Quadrat, and species initials. Write your own prompt first. Use the one below if you're stuck.

Prompt → *(use this if you're stuck)*

```
Using `Site_cleaned`, `Location_cleaned`, `Quadrat_cleaned`, and
`Species_cleaned`,
create a new column `Plant_ID`. Format example: Sa_L1_Q1_LS (Site = A, Location
= 1, quadrat = 1, Leptospermum scoparium)
Return the only the new column.
```

Step 4: Validate — sort by `Plant_ID` in Excel; check all values are unique

Step 5: Log

3.3 Notes — your turn

Prompt →

```
I am about to analyse `Plant_Age_Months`, `Height_cm`, and `Condition` for these
plants. For each row, I want to know whether the `Notes` field gives any reason
this row's
measurements might be unreliable or need special handling.
```

```
Return two new columns: `Data_Quality_Flag`: Y or N; and `Data_Quality_Notes`: a
short reason for any Y.
```

Prompt →

```
Provide me a row-by-row thematic analysis of the `Notes` field, limit your
analysis of each comment into one of three, short distinct themes from the
perspective of a botanist that you will decide. Include the analysis as column
`Notes_themes`.
```

Step 5: Log

3.5 Apply to Site B

Open a fresh Copilot conversation. Work from your Word log — apply each saved prompt to Site B.

Save combined output as `plants_combined.csv`

Segment 4 — Wrap-up

4.1 Scale test

New Copilot conversation. Drag and drop `plants_sites_AB_large.csv`.

Prompt →

```
Here is a dataset of 1,520 plant survey records from two sites.
> What I want you to do is to complete all of the same steps you completed for
site_A and _B. Return a table of all original columns (even those not modified)
and `[column_name]_cleaned` for any new, cleaned ones.

> For numerical values, flag any potential outliers with "check"; for
categorical variables, change spelling, separation errors, and flag any values
you are uncertain about with "check".

> Perform these steps:

> Cleaned or new columns      Modification prompt
Height_cm    Look at `Height_cm` only. Return a two-column table with the
original column (`Hieght_cm`), and a new column (`Height_cm_cleaned`) with
potential outlier values marked with "check".
Num_Leaves
Num_Flowers  Provide me the same summary stats and outlier checks for
`Num_Leaves` and `Num_Flowers`.

> Notes_themes  Provide me a row-by-row thematic analysis of the `Notes` field,
limit your analysis of each comment into on of three, short distinct themes from
the perspective of a botanist that you will decide. Include the analysis as
column `Notes_themes`.

> Site`Location`, `Quadrat`, `Species` and `Flowering`. Clean categorical
columns `Site`, `Location`, `Quadrat`, `Species` and `Flowering`.
I want them to be consistently spelt, all values lower-case, and values
separated by underscores "_".
Return pairs of columns for each variable: the original and a new column
`[column_name]_cleaned` for
each. In the cleaned column have the new, corrected value and use "check" if
you are unsure.

> Using `Site_cleaned`, `Location_cleaned`, `Quadrat_cleaned`, and
`Species_cleaned`,
create a new column `Plant_ID`. Format example: Sa_L1_Q1_LS (Site = A, Location
= 1, quadrat = 1, Leptospermum scoparium)
Return the only the new column.

> Plant_Age_Months  Look at `Date_Planted` and `Date_Measured`. These columns
contain dates
in inconsistent formats – some DD/MM/YYYY, some D/M/YY, some DD-Mon
with no year, and possibly some US-format (M/D/YY).
Without standardising the formats first, calculate the age of each plant
in whole months at the time of measurement. Add this as a new column
```

```
called `Plant_Age_Months`. Return the these three columns as a table.  
Flag any row where you cannot confidently interpret the dates with "check".  
  
> Condition For `Condition`, return the original column and a new column  
`Condition_cleaned` with suggested categories representing a consistent five-  
point scale from poor to excellent.  
  
> Data_Quality_Flag  
Data_Quality_Notes I am about to analyse 'Age_Months`, `Height_cm`, and  
`Condition` for these  
plants. For each row, I want to know whether the `Notes` field give any reason  
this row's  
measurements might be unreliable or need special handling.  
Add two new columns: `Data_Quality_Flag`: Y or N; and `Data_Quality_Notes`: a  
short reason for any Y."
```

What didn't work in Copilot here?

4.2 Copilot-assisted code

Prompt →

```
Produce the R code to run these steps myself in one block of code (one script) I  
can copy and paste across in one go. Put some thought into this.
```

4.3 AI-assisted analysis

Google Colab accesses the AI directly, without having to talk to it through a chatbot.

The upside of this is you can ask the AI to do much larger and more complex tasks.

For an example of all the steps done in this workshop on the large (1,520 row data set), see here:

[Data Curation Workshop - Colab Script](#)